

Khadija Hanif

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ACADEMIC BACKGROUND

University of Engineering and Technology, Lahore

2024 – Present

Bachelor of Science in Data Science | GPA: 3.563 / 4.00

Relevant Courses: Artificial Intelligence, Programming Fundamentals, OOP, Data Structures & Algorithms, Database Systems

INDUSTRIAL EXPERIENCE

Data Science / AI Intern | Oriental Nippon Development

2026 | Lahore, Pakistan

AI Job Matching Platform | 2026

Python | FastAPI | PostgreSQL | Supabase | AI Matching & Scoring | Explainable AI | TypeScript | Docker | Git

- Developed and maintained Candidate and Matching services for an AI-powered job matching platform, implementing compatibility scoring and structured match explanations.
- Designed matching outputs to expose score components and contributing factors, supporting more interpretable analysis of candidate–job compatibility.
- Diagnosed and resolved integration issues across Candidate, Matching, Employer, and Parsing services, including API validation, database schemas, structured data storage, and inter-service communication.
- Performed end-to-end debugging and testing of interconnected backend workflows and collaborated with team members to integrate and troubleshoot service components.

INDEPENDENT RESEARCH PROJECT

LLM Agent Trace & Reliability Analyzer | 2026

Python | Pandas | Pytest | SQLite | Streamlit | Agent Evaluation | Docker

- Developed a structured observability and tracing framework for tool-using LLM agents, capturing step-level decisions, tool calls, latency, retries, errors, and execution trajectories in SQLite.
- Designed an automated ground-truth benchmark and experiment pipeline to evaluate agent reliability, tool-selection behavior, execution latency, and failure modes across 17 evaluation cases and 82 recorded agent trajectories.
- Achieved a 95.12% task success rate with 62.87 ms average latency; quantified 54.92% tool-selection accuracy and identified MAX_STEPS_EXCEEDED as the primary observed failure mode.
- Built a Pandas-based reliability analysis pipeline and interactive Streamlit dashboard for trace-level debugging, failure taxonomy analysis, and visualization of agent execution behavior.

ACADEMIC PROJECTS

MedAssist — AI-Powered Medical Decision Support Prototype | 2026

Python | XGBoost | scikit-learn | SHAP | FastAPI | SQL | React

- Developed an AI-assisted medical decision-support prototype for symptom-based disease prediction and medical workflow support.
- Built and evaluated a machine-learning disease classification model achieving 98.97% test accuracy and 98.64% out-of-bag (OOB) accuracy across 41 disease classes and 134 symptom features.
- Trained the final model on 5,330 samples after data preparation and augmentation, and evaluated performance on held-out test data.
- Integrated SHAP-based explainability to analyze feature contributions underlying model predictions.
- Developed backend services and database integration for managing patient and medical workflow information, with scenario-based testing covering 8 clinical cases.

Exam Score Prediction & Smart Student Monitoring System | 2025

Python | SQL | Flask | Streamlit

- Developed a machine-learning application for predicting student examination performance using 1,000 student records and 10 feature variables.
- Compared model performance using standard regression metrics and identified Random Forest Regressor as the best-performing model.
- Achieved an R^2 score of 0.9882, with RMSE of 2.08 and MAE of 1.61 on the test set.
- Performed data preprocessing, feature engineering, model training, and evaluation using an 80/20 train-test split.
- Built an interactive Streamlit interface for generating and exploring model predictions.

RESEARCH INTERESTS

- Machine Learning
- Artificial Intelligence
- Explainable AI
- Trustworthy AI
- LLM Evaluation
- Intelligent Systems
- AI Reliability
- NLP

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

Machine Learning: scikit-learn, XGBoost, Model Evaluation, Feature Engineering

AI & Explainability: LLMs, LLM Agents, LLM Evaluation, Reliability Testing, SHAP

Backend & Applications: FastAPI, Flask, Streamlit

Testing & Engineering: Pytest, Unit Testing, Debugging, Error Handling

Databases & Tools: SQLite, MySQL, Git, GitHub, Docker, VS Code, Jupyter Notebook

CERTIFICATIONS

Google AI Essentials — Google

Intermediate SQL — DataCamp